

BAY CITY, TX, USA

WMO: 720273

Lat: 28.973N

Lon: 95.863W

Elev: 14

StdP: 101.16

Time Zone: -6.00 (NAC)

Period: 04-19

WBAN: 12981

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	0.0	2.0	-7.6	2.0	5.5	-4.8	2.5	8.5	10.9	15.2	9.8	15.2	2.4	350	0.397

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	9.8	35.7	25.8	34.8	25.9	33.8	25.9	27.9	31.3	27.5	31.1	27.1	30.7	4.4	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
27.3	23.2	29.0	26.8	22.4	28.7	26.3	21.8	28.4	89.1	31.4	87.0	31.2	85.2	30.6	32.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
9.3	8.4	7.5	-3.5	37.7	3.1	1.7	-5.7	38.9	-7.5	39.9	-9.2	40.8	-11.4	42.1		
			-4.5	28.5	2.7	1.4	-6.5	29.5	-8.1	30.3	-9.6	31.0	-11.6	32.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	21.6	12.5	15.1	18.2	21.2	24.9	28.1	28.6	28.9	26.8	22.8	17.6	14.1
	DBStd	6.84	5.01	5.26	4.40	3.52	2.62	1.50	1.49	1.71	2.11	3.81	4.99	5.22
	HDD10.0	74	30	16	3	1	0	0	0	0	0	0	4	20
	HDD18.3	612	189	113	58	14	1	0	0	0	0	10	73	153
	CDD10.0	4311	108	159	259	337	463	543	577	586	505	396	233	146
	CDD18.3	1807	10	23	55	101	205	293	318	327	255	148	52	20
	CDH23.3	17707	31	71	222	648	1818	3302	3656	3784	2427	1321	338	89
	CDH26.7	7295	1	2	14	94	555	1496	1729	1852	1041	449	58	3
Wind	WSAvg	3.1	3.5	3.8	3.9	4.0	3.6	3.0	2.4	2.3	2.3	2.7	3.0	3.3
Precipitation	PrecAvg	1199	99	64	79	84	97	120	118	86	146	113	110	88
	PrecMax	1675	273	216	236	212	212	391	464	207	428	286	297	212
	PrecMin	668	3	10	16	4	0	14	14	17	31	0	11	23
	PrecStd	290	68	54	53	58	64	83	108	56	110	80	76	56
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	26.0	27.0	28.0	30.5	32.9	36.7	36.1	37.3	36.2	32.8	30.1	27.2
		MCWB	18.5	19.7	20.8	21.6	24.0	24.9	25.8	25.5	25.8	25.3	23.4	21.8
	2%	DB	24.1	25.2	26.9	28.7	31.7	34.5	35.0	36.0	34.3	31.7	28.3	25.7
		MCWB	19.5	20.2	21.1	22.4	24.3	25.5	25.8	25.9	25.3	24.8	22.6	21.4
	5%	DB	22.6	24.1	25.8	27.7	30.7	33.4	34.1	35.0	32.9	30.7	27.0	24.1
		MCWB	19.2	19.8	20.9	22.2	24.2	25.8	26.0	26.2	25.4	24.1	21.9	21.1
	10%	DB	21.2	22.7	24.7	26.7	29.8	32.5	33.2	33.9	32.0	29.4	25.3	22.6
		MCWB	18.3	19.4	20.3	21.8	24.1	25.8	26.0	26.2	25.2	23.3	21.0	20.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	22.8	23.0	23.4	25.4	26.5	28.2	28.2	28.6	28.0	27.5	25.3	24.0
		MCDB	23.3	24.7	25.6	27.4	29.0	31.4	30.9	31.8	31.7	30.1	27.3	25.0
	2%	WB	21.5	22.3	22.8	24.7	26.0	27.5	27.7	28.0	27.4	26.6	24.4	23.0
		MCDB	22.6	23.9	24.7	26.6	28.8	30.6	30.9	31.6	31.1	29.2	26.3	24.5
	5%	WB	20.5	21.7	22.3	24.1	25.6	27.0	27.3	27.5	26.8	25.7	23.5	22.2
		MCDB	21.8	23.1	24.1	26.0	28.4	30.4	30.8	31.5	30.5	28.6	25.5	23.7
	10%	WB	19.2	20.8	21.7	23.5	25.2	26.6	26.9	26.9	26.3	25.0	22.6	21.0
		MCDB	20.8	22.0	23.5	25.4	28.0	30.2	30.6	31.0	29.9	27.9	24.7	22.1
Mean Daily Temperature Range	5% DB	MDBR	11.2	10.1	10.1	9.9	9.1	8.9	9.0	9.8	9.8	11.4	11.3	10.6
		MCDBR	12.0	9.8	9.3	9.6	9.3	10.2	10.2	11.1	11.0	10.9	11.3	10.9
		MCWBR	8.2	6.3	4.9	4.9	3.5	3.2	3.0	3.2	4.0	4.7	6.3	7.6
	5% WB	MCDBR	9.9	8.2	7.5	7.2	7.6	8.4	8.8	9.9	9.2	9.0	9.1	10.0
		MCWBR	8.4	6.1	4.5	4.2	3.3	3.2	3.1	3.5	3.8	4.9	6.4	8.1
Clear-Sky Solar Irradiance	taub	0.350	0.363	0.381	0.439	0.489	0.462	0.489	0.474	0.454	0.387	0.379	0.354	
	taud	2.487	2.460	2.398	2.235	2.126	2.267	2.207	2.251	2.276	2.452	2.420	2.483	
	Ebn at Noon	887	905	908	860	815	833	809	820	825	867	845	863	
	Edh at Noon	93	103	116	141	158	137	145	138	130	103	98	89	
All-Sky Solar Radiation	RadAvg	2.90	3.34	4.52	5.40	6.08	6.47	6.28	6.00	5.09	4.44	3.34	2.61	
	RadStd	0.30	0.55	0.40	0.45	0.41	0.44	0.58	0.39	0.50	0.47	0.35	0.32	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (34 neighbors)	+0.33	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+115	+99

Nomenclature: See separate page